Behavioural analysis of Marine Protected Areas(MPAs) design and expansion optimization for bio-diversity conservation

A PROJECT REPORT

Submitted to

Amrita Vishwa Vidyapeetham

in partial fulfillment for the award of the degree of

Bachelor of Technology in Computer Science and Engineering (AIE)

By

NEELRAJ R

(CH.EN.U4AIE22035)

SHYAM PRASATH J

(CH.EN.U4AIE22051)

Supervisor

Dr. R Prasanna Kumar

(Associate Professor, AIE Department, ASE)

AMRITA VISHWA VIDYAPEETHAM

AMRITA SCHOOL OF COMPUTING

CHENNAI – 601103

October 2025

BONAFIDE CERTIFICATE

Certified that this project report **“Behavioural analysis of Marine Protected Areas(MPAs) design and expansion optimization for bio-diversity conservation”** is the bonafide work of **“Mr. Neelraj R & Mr. Shyam Prasath J”** who carried out the project work under my supervision.

SIGNATURE	**SIGNATURE**
Dr. S. SOUNTHARRAJAN	**Dr. R PRASANNA KUMAR**
CHAIRPERSON	**SUPERVISOR**
Department of CSE.	Department of CSE(AIE).
Amrita School of Computing	Amrita School of Computing
Chennai	Chennai

INTERNAL EXAMINER **EXTERNAL EXAMINER**

DECLARATION BY THE CANDIDATE

I declare that the report entitled **"Behavioural analysis of Marine Protected Areas(MPAs) design and expansion optimization for bio-diversity conservation"** submitted by me for the degree of Bachelor of Engineering is the record of the project work carried out by me under the guidance of **"Dr. R PRASANNA KUMAR"** and this work has not formed the basis for the award of any degree, diploma, associateship, fellowship, titled in this or any other University or other similar institution of higher learning.

SIGNATURE

NEELRAJ R
(CH.EN.U4AIE22035)
Computer Science and Engineering (AIE)
Amrita School of Computing
Chennai.

ABSTRACT

Marine Protected Areas (MPAs) are essential for conserving marine biodiversity and supporting sustainable fisheries. As nations strive toward the “30 by 30” goal of protecting 30% of the ocean by 2030, the design and management of MPA networks present a complex challenge that must balance ecological effectiveness with socio-economic realities. To address this, this project details a high-performance agent-based model (ABM) that serves as a virtual laboratory for simulating these dynamics and evaluating management strategies under realistic pressures. Our spatially explicit framework models a fish population whose life cycle, growth (von Bertalanffy), and movement are governed by a multi-layered environment, including procedurally generated bathymetry, seasonal temperatures, and a dynamic habitat quality grid. The simulation directly assesses the dual anthropogenic impacts of fishing fleet and persistent habitat degradation emanating from multiple coastal ports. The fleet is comprised of individual, profit-driven bio-economic agents, each constrained by fuel and cargo capacity, who employ an adaptive exploit-versus-explore strategy to locate fish. The model's primary application is to assess the effectiveness of a central MPA in conserving fish stocks and supporting the fishery, despite these combined external stressors. By analyzing key performance indicators such as fish biomass, population stability, MPA spillover effects, and overall fleet profitability, this framework provides a robust tool for testing the resilience of fishery management policies in complex, real-world scenarios.

Keywords: Agent-Based Model, Fisheries Management, Marine Protected Area (MPA), Socio-Ecological System, Fisher Behavior, Habitat Degradation, Bio-economics, Spatio-Temporal Dynamics

ACKNOWLEDGEMENT

This project work would not have been possible without the contribution of many people. It gives me immense pleasure to express my profound gratitude to our honorable Chancellor **Sri Mata Amritanandamayi Devi,** for her blessings and for being a source of inspiration. I am indebted to extend my gratitude to our Director, **Mr. I B Manikantan** Amrita School of Computing and Engineering, for facilitating us all the facilities and extended support to gain valuable education and learning experience.

I register my special thanks to **Dr. V. Jayakumar,** Principal, Amrita School of Computing and Engineering for the support given to me in the successful conduct of this project. I wish to express my sincere gratitude to my supervisor **Dr. R Prasanna Kumar,** Associate Professor, **Dr. S. Sountharrajan,** Chairperson & **Dr. S. Baghavathi Priya,** Program Chair, Department of Computer Science and Engineering, for his inspiring guidance, personal involvement and constant encouragement during the entire course of this work.

I am grateful to Project Coordinator, Review Panel Members and the entire faculty of the Department of Computer Science & Engineering, for their constructive criticisms and valuable suggestions which have been a rich source to improve the quality of this work.

Neelraj R & Shyam Prasath J

(CH.EN.U4AIE22035 & CH.EN.U4AIE22051)

TABLE OF CONTENTS

CH. NO.	TITLE	PAGE
	Abstract	iv
	List of Tables	vii
	List of Figures	viii
	List of Symbols and Abbreviations	ix
1	INTRODUCTION	1
	1.1 BACKGROUND	1
	1.2 CHALLENGES FACED BY FARMERS	2
	1.3 MOTIVATION	2
	1.4 OBJECTIVES OF THE PROJECT	3
2	LITERATURE REVIEW	5
3	METHODOLOGY	9
	3.1 SIMULATION CORE	9
	3.2 DYNAMIC AND AUTOMATED ENVIRONMENT	10
	3.3 REWARD FUNCTION	15
	3.4 AGENT TRAINING	16
4	RESULTS	19
5	CONCLUSION AND FUTURE WORK	27
	5.1 CONCLUSION	27
	5.2 FUTURE WORK	27
	REFERENCES	29

LIST OF TABLES

TABLE NO.	TITLE	PAGE
4.1	Description of parameters	10

LIST OF FIGURES

FIGURE NO.	TITLE	PAGE
1.1	Overview of Workflow	4
3.1	The structure and visualization of how environment is simulated and how RL agent trains upon that. Well documented configuration files control crop and soil dynamics	11
3.2	Soft Actor Critic Framework that learns irrigation task	17
4.1	Analysis of the trained agent’s adaptive irrigation policy (Under heavy rain)	19
4.2	Analysis of the trained agent’s adaptive irrigation policy (Under moderate rain)	22
4.3	Analysis of the trained agent’s adaptive irrigation policy (Under low rain)	24

LIST OF SYMBOLS AND ABBREVIATIONS

DRL	-	Deep Reinforcement Learning
CNN	-	Convolutional Neural Network
SAC	-	Soft Actor Critic

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

Marine ecosystems play a crucial role in maintaining global biodiversity, regulating climate, and supporting the livelihoods of millions of people. However, overfishing, habitat degradation, and climate change have increasingly threatened the stability and productivity of ocean systems. To counter these pressures, Marine Protected Areas (MPAs) have emerged as a key strategy for conserving marine biodiversity and ensuring sustainable fisheries. MPAs are geographically defined zones where human activities are regulated to protect ecological integrity. They can be fully protected "no-take" zones, where all extractive activities are prohibited, or partially protected multi-use zones, which allow limited and regulated access.

The global conservation community, under the *Kunming–Montreal Global Biodiversity Framework*, has set an ambitious target to protect 30% of the world's oceans by 2030—popularly known as the "30 by 30" goal. While MPAs have been shown to enhance fish biomass, promote ecosystem recovery, and even increase fisheries productivity through spillover effects, their implementation often presents social and economic challenges. In some regions, restricted access to traditional fishing grounds has led to loss of income, food insecurity, and community conflicts. Moreover, weak enforcement and inadequate management frequently render many MPAs ineffective.

Designing effective MPA networks thus requires balancing ecological gains with socio-economic sustainability. Previous studies have primarily focused on spatial optimization and ecological metrics, often overlooking critical human dimensions such as fisher cooperation and behavioral adaptation. Additionally, few studies have incorporated food-web interactions that determine broader ecosystem stability. Addressing these gaps is vital to develop holistic conservation strategies that account for both ecological complexity and human decision-making. Understanding how cooperative behaviors among fishers interact with ecological processes within MPAs can lead to more resilient and adaptive management frameworks that support both biodiversity conservation and sustainable livelihoods.

1.2 CONSERVATION-FISHERY TRADE-OFFSC

Marine life is under increasing pressure from intensive fishing activities and inadequate spatial management. Overexploitation has reduced fish stocks and disrupted food-web dynamics, making recovery difficult without targeted conservation measures such as Marine Protected Areas (MPAs). However, poorly designed or weakly enforced MPAs often fail to deliver their intended ecological benefits. Many exist only on paper, lacking sufficient monitoring, enforcement, or community participation. As a result, illegal or displaced fishing around MPA boundaries can lead to localized overfishing, further stressing vulnerable species and habitats.

Socio-economic conflicts also pose significant challenges. When MPAs restrict access to traditional fishing grounds, they can cause income losses and resentment among coastal communities, leading to non-compliance and management failure. Furthermore, most existing models for MPA design focus primarily on ecological parameters—such as size, spacing, and habitat representation—while overlooking the behavioral aspects of fishers. The absence of cooperative fishing behavior and collective management can undermine long-term sustainability. Additionally, limited consideration of food-web interactions constrains understanding of how MPAs influence predator–prey relationships and ecosystem stability. These challenges underscore the need for integrative modeling approaches that couple ecological, behavioral, and management dimensions to achieve balanced and effective marine conservation outcomes.

1.3 MOTIVATION

The rapid global expansion of Marine Protected Areas (MPAs) under the “30 by 30” biodiversity target highlights the urgent need for effective and equitable marine conservation strategies. While MPAs have demonstrated ecological success in restoring fish stocks and protecting habitats, their long-term effectiveness often depends on human behavior and management design. Most existing studies focus on ecological optimization—such as MPA size, spacing, and habitat coverage—while neglecting critical socio-behavioral dimensions like fisher cooperation and compliance. This oversight can lead to policies that appear successful on paper but fail in real-world application due to non-cooperative or displaced fishing activities.

Moreover, traditional modeling approaches rarely integrate ecosystem complexity, particularly food-web interactions that determine overall ecological stability. The lack of socio-economic cost considerations and advanced computational approaches further limits our ability to design adaptive and sustainable MPA networks. This project is motivated by the need to bridge these gaps through an **agent-based modeling (ABM)** framework that captures the coupled dynamics of fish populations, fisher behavior, and ecosystem interactions. By leveraging simulation tools and emerging methods like reinforcement learning, the study aims to create more realistic and effective MPA designs that balance biodiversity conservation with human livelihoods.

1.4 OBJECTIVES OF THE PROJECT

1. To construct a high-fidelity agent-based simulation framework that acts as a comprehensive **testbed for Marine Protected Area (MPA) analysis**. This framework simulates the complete life history and spatio-temporal dynamics of a fish population, including individual growth (von Bertalanffy), population renewal (Beverton-Holt), and complex movement (combined taxis) in response to a dynamic, multi-layered marine environment featuring procedurally generated bathymetry, seasonal temperatures, and an evolving habitat quality layer.
2. To integrate and simulate the dual anthropogenic pressures that define the realistic socio-ecological conflict within the testbed. This involves modeling the direct pressure of resource extraction from a non-cooperative, bio-economic fishing fleet operating under operational constraints, alongside the indirect pressure of persistent habitat degradation emanating from procedurally generated coastal port infrastructure.
3. To utilize the simulation framework to evaluate the efficacy of a static, central MPA under these combined pressures, thereby establishing a robust analytical baseline. This evaluation, based on a comprehensive suite of ecological and socio-economic indicators, is designed to provide the foundational insights necessary to guide future **MPA expansion and network optimization studies** within this virtual environment.

CHAPTER 2

LITERATURE REVIEW

The design and expansion of Marine Protected Area (MPA) networks represent a critical tool in global biodiversity conservation, yet their effectiveness hinges on navigating a complex interplay of ecological processes, economic pressures, and human behavior. While the goal of protecting vast ocean areas is clear, the methodology for achieving this optimally is an area of intense research. The literature reveals a progression from static, spatially-focused optimization to more dynamic, integrated socio-ecological frameworks. However, significant gaps persist in unifying advanced optimization techniques with realistic models of fleet dynamics, fisher cooperation, and broader ecosystem-wide impacts. This review surveys the state-of-the-art across these domains to identify the critical research questions that remain unanswered.

The foundation of modern MPA design lies in Systematic Conservation Planning (SCP), which has been operationalized through powerful computational tools. A key example is the prioritizr R package, which utilizes Integer Linear Programming (ILP) and exact solvers to identify optimal reserve configurations that meet specific conservation targets at a minimum cost [13]. This approach has proven scalable, enabling basin-wide planning that incorporates diverse features from vulnerable marine ecosystems to future habitat refugia [2]. The sophistication of these optimization models has advanced beyond simple site selection to address the spatial quality of the resulting network. Researchers have employed Mixed-Integer Linear Programming (MILP) with multicommodity-flow formulations to explicitly design reserves that are compact, connected, and free of internal gaps, which are attributes thought to enhance ecological resilience and manageability [12]. Similarly, heuristic methods like Simulated Annealing have, in some regional contexts, outperformed Multi-Criteria Evaluation (MCE) by yielding networks with superior compactness and contiguity, demonstrating the importance of algorithm choice for achieving specific spatial goals [9]. As the scale of these problems has grown to encompass entire oceans, the computational burden has become a major obstacle. To address this, graph-based clustering methods have been proposed to partition massive problems into smaller, tractable sub-problems, allowing for near-optimal solutions on modest hardware without sacrificing essential connectivity goals [10]. The ecological realism of these plans is also being enhanced by moving from two-dimensional to three-dimensional prioritization, with tools like prior3D using depth-stratified optimization to account for vertical biodiversity turnover in marine ecosystems [14]. A multi-objective ILP model using the AUGMECON2 method has also shown promise in delivering diverse and highly compact zoning solutions, outperforming traditional weighted-sum approaches in solution quality and relevance [5]. While these optimization tools provide a robust mathematical foundation for spatial planning, they often treat socio-economic factors as static cost layers, failing to capture the dynamic behavioral responses of the human systems they are intended to manage.

A primary limitation of purely spatial optimization is its inability to account for the adaptive behavior of fishing fleets. The implementation of an MPA does not occur in a vacuum; global-scale analysis confirms that while fishing activity decreases inside protected zones, this effort is largely displaced to adjacent areas, often intensifying pressure at MPA boundaries [20]. This realization has spurred the development of complex socio-ecological models designed to simulate these human-natural system interactions. Agent-Based Models (ABMs) have emerged as a particularly effective tool for this purpose, capable of simulating spatially explicit fleet dynamics and stock interactions under various management scenarios [16]. These models can reveal critical trade-offs; for instance, closing 20% of high-density fishing grounds may increase stock biomass by 28%, but it could also lead to a 12% drop in fleet profits and encourage risk-prone fishers to overexploit the remaining areas [16]. The structural fidelity of these models is paramount. A direct comparison showed that "low-definition" surplus production models systematically overestimated the biomass benefits of MPAs in 91% of simulations when compared to a "high-definition," age-structured, multi-fleet model like marlin, which better captures the negative effects of effort displacement and habitat heterogeneity [3, 15]. The human dimension is further complicated by external economic forces, as models show that market stability can paradoxically stimulate over-investment and concentrated harvesting by risk-seeking fishers, demonstrating that market dynamics can significantly alter ecological outcomes [4]. Perhaps the most crucial, and often overlooked, behavioral factor is cooperation. An ABM designed to test this variable found that cooperation among fishers improved overall stock biomass by

22% compared to non-cooperative scenarios, suggesting that the social fabric of a fishing community can be as determinative of MPA success as its size or placement [18]. The clear gap identified in this body of work is the disconnect between these sophisticated behavioral simulations and the optimization frameworks used for initial MPA design.

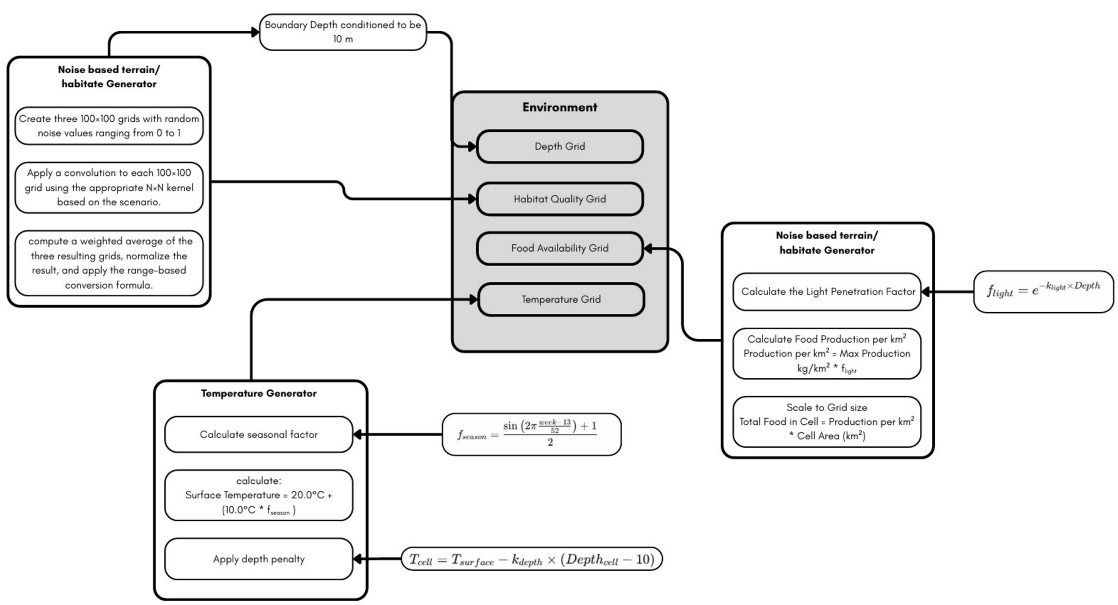
To address the high dimensionality and non-linearity inherent in these socio-ecological systems, researchers are increasingly turning to advanced computational paradigms like Reinforcement Learning (RL). Unlike traditional optimization, which solves for a single static solution, RL agents can learn adaptive management policies through interaction with a simulated environment. This approach has shown great potential, with Deep Reinforcement Learning (DRL) discovering near-optimal fishery harvest quotas that were more responsive to stock depletion than historical practices [1]. In a spatial context, DRL agents have been successfully trained to optimize for ecological connectivity across large landscapes, demonstrating an ability to derive complex spatial strategies that may not be intuitive to human planners [11]. This paves the way for a shift from static to dynamic management. For example, work using Approximate Dynamic Programming (ADP) has shown that Dynamic Spatial Closures (DSCs), which open and close based on real-time conditions, can outperform fixed MPAs by producing higher harvests and greater habitat protection, especially where strong habitat-stock linkages exist [6]. However, these cutting-edge methods are not without challenges. They are computationally demanding, their convergence is sensitive to hyperparameter tuning, and the resulting policies can be opaque, posing a risk if the reward functions are misspecified or fail to capture key system objectives [1, 11].

Ultimately, the efficacy of an MPA network must be evaluated from a holistic, ecosystem-based perspective that extends beyond single species or sectors. Ecological connectivity is a cornerstone of marine ecosystem function, and planning workflows are now being developed that explicitly combine connectivity modeling, using techniques like least-cost path analysis and circuit theory, to delineate ecological corridors that enhance network coherence [8]. Moreover, the trophic interactions within an ecosystem can lead to cascading effects that are invisible to single-species models. Spatially explicit food web models have demonstrated that MPAs can significantly improve the biomass of protected species, which in turn can increase predator abundance by 12-18%, highlighting an important ecosystem-level benefit [19]. This ecosystem view must also encompass the full spectrum of human activities. Cross-sectoral spatial planning, which integrates the opportunity costs of industries like fishing, shipping, and mining, has been shown to achieve conservation targets with dramatically lower economic impact than sector-specific plans [7]. For instance, a cross-sectoral approach reduced projected fishing revenue loss from 54% to a mere 2% [7]. Even models designed for other industries, such as offshore energy development, confirm that spatial closures can yield significant conservation co-benefits by reducing habitat disturbance and allowing for biomass gains in species with low mobility [17]. In synthesis, the literature points toward a clear need for a unified framework that bridges these disparate research streams. While powerful tools exist for spatial optimization, behavioral modeling, advanced computation, and ecosystem analysis, they have largely evolved in isolation. The critical research gap lies in the integration of these components: a framework that can optimize MPA design while explicitly accounting for the dynamic, cooperative, and non-cooperative behaviors of fishing fleets, and simultaneously evaluating the cascading impacts of these interactions on the broader food web.

CHAPTER 3

METHODOLOGY

This study utilizes a spatially explicit agent-based model (ABM) to investigate the complex, emergent dynamics of a socio-ecological fishery system. The model is designed as a virtual laboratory to simulate the interactions between a biological population (fish), economic actors (a fishing fleet), and a dynamic marine environment. The core objective is to evaluate the efficacy of a static Marine Protected Area (MPA) as a conservation and management tool under the combined, simultaneous pressures of resource extraction and persistent, localized anthropogenic habitat degradation. The model's architecture enables large-scale simulations involving tens of thousands of individual agents, facilitating the analysis of system-level properties that emerge from micro-level behaviors and interactions.



3.1 The Simulation Domain: A Dynamic and Heterogeneous Seascape

The model operates within a continuous, two-dimensional domain representing a 90,000 square kilometer marine area, structured as a 300 km by 300 km cartesian plane. This domain is not an inert backdrop but a multi-layered, dynamic environment that directly influences the behavior and fate of the agents within it. The simulation progresses in discrete weekly time steps over a configurable multi-year period, allowing for the observation of both short-term behaviors and long-term ecological and economic outcomes.

- **Bathymetric and Geomorphological Layer:** The foundation of the virtual world is a static bathymetric grid that defines the seafloor topography. This layer is procedurally generated at the simulation's outset using a sophisticated multi-layered noise algorithm. By convolving several noise fields at different spatial frequencies and amplitudes, the model creates a complex and realistic underwater landscape, complete with shallow coastal shelves, deeper offshore basins, plateaus, and trenches. This geomorphological structure is a critical static element, as it directly influences the spatial distribution of other environmental variables, most notably temperature and primary productivity, which are depth-dependent.
- **Hydrodynamic and Thermal Layer:** The thermal environment is a dynamic grid layer that is recalculated at every weekly time step. It simulates two key oceanographic features. First, it models seasonal temperature fluctuations via a sinusoidal function, creating warmer surface waters in summer months and cooler temperatures in winter. Second, it incorporates a vertical temperature gradient, where water temperature decreases as a function of depth from the seasonally-adjusted surface temperature. This creates a complex, spatio-temporally variable thermal landscape that directly impacts the metabolic processes and habitat preferences of the fish agents.
- **Ecological and Habitat Quality Layer:** A crucial dynamic layer represents the overall habitat quality, a composite index of ecosystem health that influences fish movement and reproductive success. Initialized with a heterogeneous spatial pattern, this grid evolves weekly through two primary feedback mechanisms:
    - **Trophic Feedback (Bottom-Up Control):** Habitat quality is linked to the balance between local primary production and consumption. Primary production is modeled as a function of depth, simulating light attenuation and thus higher food availability in shallower waters. The model calculates the total fish biomass in each grid cell and the corresponding food consumption. If consumption exceeds local production, the habitat quality degrades; conversely, it recovers in areas of low grazing pressure. This creates a direct

feedback loop between the fish population's density and the health of its environment.

  - **Anthropogenic Feedback (Pollution Pressure):** The model simulates persistent, localized pollution from static port agents located on the coastal boundaries. Each port projects a continuous, negative pressure on the adjacent habitat. This impact is modeled using an exponential decay function, where the degradation is most severe at the shoreline source and diminishes with distance into the sea. This creates permanent zones of compromised habitat quality that fish are behaviorally incentivized to avoid and where reproductive success is likely diminished.

3.2 Quantitative Formulations and Ecological Processes

The agent behaviors and environmental interactions are governed by a set of deterministic and stochastic mathematical formulations derived from established ecological theory.

- Individual Fish Growth Model: The growth of each fish agent is a two-step process. First, the length of a fish at a given age is determined by the von Bertalanffy Growth Function. This is a widely used equation in fisheries science where the length is calculated as a function of the species' asymptotic maximum length, a growth coefficient determining how quickly it reaches that maximum, and the theoretical age at which its length would be zero. The formula incorporates an exponential term to ensure that growth is rapid in early life and slows as the fish approaches its maximum size. Second, the biomass of the fish is calculated from its length using a non-linear allometric power law. In this relationship, the weight of the fish is proportional to its length raised to an exponent, a value typically close to three for most fish species.

$$L_t = L_\infty(1 - e^{-K(t-t_0)})$$

L_t : This is the **length of the fish at age t**. It is the output of the equation.

L_∞ : This is the **asymptotic maximum length**. It represents the theoretical maximum size the fish would reach if it lived forever.

K : This is the **growth coefficient**. It controls how quickly the fish approaches its maximum length. A larger value means faster growth.

t : This is the **current age of the fish** in years.

t_0 : This is the **theoretical age at which the fish would have had zero length**. It is a mathematical correction factor to ensure the curve accurately fits real-world growth data, especially for the earliest life stages.

- Population Stock-Recruitment Model: The number of new fish entering the population each year is governed by the Beverton-Holt stock-recruitment model. This function calculates the number of new recruits based on the total biomass of the adult spawning stock. The relationship is formulated such that at low stock densities, the number of recruits is nearly proportional to the spawning biomass. However, the equation includes a density-dependent term in its denominator that causes the recruitment rate to level off as the spawning stock becomes very large, representing the effects of limited resources and increased competition on the survival of early life stages.

$$R = \frac{\alpha \cdot S}{1+\beta \cdot S}$$

R : This is the **number of new recruits** that successfully enter the population. This is the value the equation calculates.

α : This is the parameter representing the **maximum recruitment rate at very low stock densities**. You can think of it as the number of surviving recruits produced per kilogram of spawning stock when the population is very small and there is no competition.

S : This is the **Spawning Stock Biomass (SSB)**. It is the total weight (in kilograms) of all the adult fish in the population that are capable of reproducing.

β : This is the **density-dependent parameter**. It controls how quickly the recruitment rate levels off as the spawning stock increases. A larger beta means that competition and other density-dependent factors (like limited nursery habitat or increased predation on larvae) become significant at lower stock sizes.

- Combined Taxis Movement Model: The final direction of movement for each fish agent is calculated as a weighted sum of three distinct behavioral drives. The final movement vector is composed of a random component representing stochastic exploration, a density-dependent component directing the fish away from the center of its current high-density grid cell, and an environmental component directing the fish towards the neighboring cell with the highest habitat suitability score. The relative importance of each of these three drives is controlled by a set of weights, allowing the model to be tuned to represent species with different behavioral tendencies.

$$\theta_{final} = (\omega_r \cdot \theta_{rand}) + (\omega_d \cdot \theta_{dens}) + (\omega_e \cdot \theta_{env})$$

θ_{final} : This is the **final angle of movement** for the fish in radians. The fish will move a set distance in this direction.

ω_r : This is the **weight given to the random component** of movement, representing stochastic, exploratory behavior.

θ_{rand}: This is a **randomly generated angle** between 0 and 2π radians, providing the unpredictable element of the fish's movement.

ω_d : This is the **weight given to the density-dependent component**, representing the fish's tendency to avoid overcrowded areas.

θ_{dens} : This is the **repulsion angle**. It is calculated to point directly away from the center of the current grid cell if the fish density in that cell is high. If the density is low, this component has no effect (it defaults to the random angle).

ω_e : This is the **weight given to the environmental component**, representing the fish's instinct to seek out better living conditions.

θ_{env} : This is the **attraction angle**. The model assesses all neighboring grid cells and calculates a "suitability score" for each based on a combination of optimal temperature, optimal depth, and habitat quality. This angle points towards the neighboring cell with the highest suitability score.

- Fisheries Harvest Equation: The quantity of fish harvested by a single vessel is modeled using a standard catch formulation. The number of fish caught is calculated as the product of three factors: the catchability coefficient, which represents the intrinsic efficiency of the fishing gear; a standardized measure of fishing effort; and the total number of legally harvestable fish available within the vessel's sensor detection radius. Legally harvestable fish are defined as those above a specified minimum size and located outside the boundaries of the Marine Protected Area.

$$C = Q \cdot e \cdot N$$

C : This is the **Catch**, specifically the number of individual fish harvested by the vessel in a given time step.

Q : This is the **catchability coefficient**. It is a crucial parameter that represents the intrinsic efficiency of the fishing gear and the skill of the fisher. It is the fraction of the available fish stock that is caught by one unit of fishing effort.

e : This is the **fishing effort**. In this model, it is a standardized value representing the intensity of the fishing activity. While it is set to a constant, in more complex models it could vary based on factors like weather or vessel power.

N: This is the **number of available, legal fish**. It represents the total count of individual fish agents that are (1) within the vessel's sonar detection radius (parameter R), (2) outside the boundaries of the Marine Protected Area, and (3) larger than the minimum legal harvest size .

The specific scenario investigated in this study is defined by a comprehensive set of parameters that configure the simulation's spatial, temporal, biological, and economic dimensions.

- **Simulation Configuration:** The model is initialized to run for a total duration of three simulated years, corresponding to 156 weekly time steps. The initial fish population is set to 100,000 individual agents. The environmental grid layers, which resolve the spatial variation in depth, temperature, and habitat quality, are configured with a resolution of 100 by 100 cells. The Marine Protected Area is centrally located and designated to cover twenty percent of the total simulation area.
- **Biological and Ecological Parameters:** The model is parameterized for a generic large-bodied fish species, *Catla*. The optimal environmental conditions sought by the fish agents are defined as a temperature of 25 degrees Celsius and a depth of 30 meters. The species' life history is structured with a larval phase of 4 weeks and a juvenile phase of 52 weeks, after which individuals are classified as adults. Weekly mortality probabilities are stage-specific, set at 2 percent for larvae, 0.1 percent for juveniles, and 0.05 percent for adults. The annual spawning season is defined to occur between the 23rd and 35th weeks of the year, with recruitment occurring at the end of this period. The constants for the growth and recruitment equations are set based on established literature values for similar species. The maximum carrying capacity of the ecosystem is set to a total of 70 million kilograms of biomass.

- **Fisheries and Economic Parameters:** The fishing fleet is initialized with 750 vessels, a number derived from the initial placement of 15 ports, each supporting 50 vessels. Each vessel operates with a fuel capacity of 600 liters and a fuel efficiency of 5 kilometers per liter, yielding a maximum operational range of 3,000 kilometers on a full tank. The cargo hold for each vessel is limited to a capacity of 300 kilograms per trip. The technical specifications of the fishing gear include a sonar search radius of 5 kilometers and a catchability coefficient of 0.7. The economic environment is defined by a fixed operational cost of 5,000 Rupees per vessel per trip, a fuel price of 90 Rupees per liter, and an average market price of 150 Rupees per kilogram for landed fish. A minimum harvestable size of 0.5 meters is enforced as a technical management measure.
- **Behavioral Parameters:** Agent decision-making is controlled by a specific set of behavioral parameters. For fish agents, the weights for the combined taxis movement model are configured to give the highest influence to random exploration (a weight of 0.7), followed by density-avoidance (0.2) and environment-seeking (0.1). For fisher agents, the "exploit" phase of their strategy is governed by a memory limit, allowing a vessel to remain at a last known successful spot for a maximum of five sub-steps before being forced to re-evaluate or explore.
- **Anthropogenic Impact Parameters:** The simulation domain features 15 ports, which are procedurally generated along the west and south boundaries at initialization. These ports are required to have a minimum spacing of 5 kilometers from each other and the domain corners. The physical size of each port along the coastline can be up to a maximum of 10 kilometers. Each port generates a persistent pollution field, with the depth of this degradation zone into the sea being one-third of the port's coastal length. The maximum rate of habitat degradation at the source of this pollution is set to 0.2 percent per week.

Multi-scalar procedural noise:

To ensure that the model is generalizable and not tied to a specific real-world geography, this environment is generated procedurally at the simulation's outset using the principle of multi-scalar procedural noise. This process creates the bathymetric map that dictates the seafloor topography and the initial habitat quality map that represents the baseline ecosystem health.

1. **Multi-Scalar Noise for Terrain Generation**
   The model's seafloor is not a uniform plane but a complex terrain with varied topography. This is achieved by generating and combining several layers of smoothed, random noise, each corresponding to a different spatial scale of geological features. The process begins by creating grids of uncorrelated random values, which are then smoothed using a two-dimensional convolution operation. The size of the averaging window, or kernel, used in this operation determines the scale of the resulting features.
   - **Layer 1: Small-Scale Features:** A small convolution kernel is applied to the first grid of random values. This minimal averaging produces fine-grained, spatially correlated noise, which represents the small-scale roughness and localized texture of the seafloor.
   - **Layer 2: Medium-Scale Features:** A medium-sized kernel is used on the second grid. This level of smoothing generates intermediate topographical elements that are larger and more defined than the small-scale roughness, akin to regional features like hills, valleys, and ridges.
   - **Layer 3: Large-Scale Features:** A large kernel is applied to the third grid. The extensive averaging from this kernel produces broad, slowly varying features. This layer represents the dominant geomorphology of the region, such as large basins, expansive plateaus, and continental slopes, forming the foundational structure of the landscape.
2. These three distinct layers of smoothed noise are then combined into a single composite grid through a weighted average.
3. **Shallow Water Buffer Enforcement**
   After the initial terrain is generated, a crucial final step is the enforcement of a realistic coastal shelf. This procedural correction ensures a plausible geomorphology near the land-sea interface where ports are located. The model identifies all grid cells within a 20-kilometer coastal zone and applies a blending function that algorithmically forces the depth to be shallow, transitioning smoothly from a maximum depth of 10 meters at the shoreline. This prevents the generation of unrealistic deep-sea trenches immediately adjacent to the coast.

3.3 Agent Formulations, Behavioral Rules, and Life Histories

- Fish Agents: The Biological Population: The biological population is composed of individual fish agents, each characterized by a set of state variables including positional coordinates, age in weeks, length, biomass, life stage (larval, juvenile, or adult), and survival status. The population's size and structure evolve through weekly processes of aging and mortality. The probability of mortality is stage-specific, with larval and juvenile stages having significantly higher baseline mortality rates than adults. The population is replenished annually through the Beverton-Holt recruitment event, with the spatial distribution of new larvae being biased towards healthier habitats.
- Fisher Agents: The Bio-Economic Fleet: The fishing fleet consists of numerous individual, non-cooperative bio-economic agents. Each vessel is an autonomous entity whose behavior is driven by the goal of maximizing profit within a set of operational constraints. Vessel movement and fishing location choice are governed by an adaptive exploit-versus-explore strategy based on economic feedback. Following a profitable trip, a vessel will engage in exploitation, returning to its last known successful fishing location. Conversely, if a trip is unprofitable or a known location becomes depleted, the agent switches to exploration, moving to a new, randomly selected location. This decision logic is constantly moderated by a fuel-safety check; a vessel will not initiate a trip if its fuel is insufficient for a round trip to the target destination and back to its home port, and it will abort its current activity to return home if fuel levels become critical or its cargo hold nears capacity.
- Port Agents: Static Infrastructure: The ports are static infrastructure agents positioned along the domain's coastal boundaries. They serve a dual role: first, as the home base for all fisher agents, functioning as the start and end point for all fishing trips where refueling and unloading of catch occurs. Second, they act as the point sources for the persistent, long-term habitat degradation that impacts the coastal ecosystem.

3.4 Rules and strategies of Agents

The Fisher Agent's Behavioral and Harvesting Rules

The behavior of each individual fisher agent is governed by a sophisticated, adaptive strategy that dictates both its movement and its harvesting actions on a weekly basis. This rule set, termed the "Adaptive Exploit-Versus-Explore Strategy," is a hierarchical process where the agent first makes a movement decision based on its current state and past success, and then attempts to harvest at its new location.

Part 1: The Movement Decision - A Hierarchical Process

The agent's first task is to determine its target coordinates for the week. This is not a single calculation but a series of prioritized checks.

1. **Overriding Safety Protocols (Return to Port):** Before considering any fishing activity, the agent performs a critical self-assessment. A mandatory return to its home port is triggered if either of two conditions is met:
   - **Low Fuel:** The vessel's current fuel level is insufficient to make a round trip to its current location plus a small safety buffer.
   - **Near-Full Cargo Hold:** The vessel's cargo hold is at or above 80% of its maximum capacity. If either of these conditions is true, the agent's only goal is to return home, and all subsequent strategic decisions are bypassed.
2. **Strategic Choice (Exploit vs. Explore):** If the safety protocols are not triggered, the agent proceeds to its primary strategic decision-making. This logic is driven by the profitability of its previous trip and a short-term memory mechanism.
   - **Exploitation Phase:** The agent's default behavior is to exploit known or nearby resources. This happens in two tiers: i. **Local Optimum Search:** The agent first uses its sonar to scan the immediate vicinity for the grid cell containing the highest biomass of legally harvestable fish. If such a "hotspot" is detected, it becomes the agent's target destination. ii. **Memory-Based Fallback:** If no superior local hotspot is found, the agent relies on its memory. If its last trip was profitable, it will target the exact coordinates of its last known successful catch.
   - **The Exploitation Limit:** This exploitation behavior is not indefinite. The agent maintains a counter for how many consecutive steps it has spent exploiting a single area. If this counter exceeds a predefined limit (the "exploit memory limit," set to 5), the agent is forced to abandon this strategy, even if it

is still profitable. This mechanism prevents the agent from getting stuck at a single, marginally profitable location and encourages a broader search.

  - **Exploration Phase:** The agent switches to an exploration strategy if one of two conditions is met: (1) its previous fishing trip was not profitable, or (2) the exploitation memory limit has been reached. In this mode, the agent calculates a new, random target location within a wide search radius of its last known successful spot. This represents a stochastic search for new, potentially more profitable fishing grounds.

3. **Final Fuel Safety Check:** Before committing to travel to any target destination other than its home port, the agent performs a final, critical fuel calculation. It checks if its current fuel supply is sufficient to travel from its current position to the new target, and then from that target all the way back to its home port. If the total fuel required for this entire prospective journey exceeds its current supply, the trip is deemed unsafe. In this case, the agent's decision is overruled, and its target is reset to its home port.

Part 2: The Harvesting Process

After the agent has moved to its final target location for the time step, it initiates the harvesting process.

1. **Detection via Sonar:** The vessel scans a circular area around its current position, defined by its sonar radius (5 kilometers), to detect all individual fish agents within range.
2. **Filtering of Targets:** The list of detected fish is then filtered based on two key criteria:
   - **Legality (MPA Compliance):** Any fish agent whose coordinates are located inside the boundaries of the Marine Protected Area is removed from the list of potential targets.
   - **Legality (Size Compliance):** Any fish agent whose length is below the minimum legal harvest size (0.5 meters) is also removed from the list.
3. **Calculating and Executing the Catch:** The final number of fish to be harvested is calculated using the Baranov catch equation $C = Q \cdot e \cdot N$, where N is the number of fish remaining after the filtering process. The model then randomly samples this number of individuals from the list of legal targets. For each harvested fish, its

is_alive status is set to False, its biomass is added to the vessel's cargo hold, and its value is added to the vessel's revenue for the current trip. If any catch is made, the vessel updates its internal memory with its current coordinates as the new "last successful fishing spot."

3.4 Model Assumptions and Strategic Simplifications

To focus the analysis on the research objectives and ensure computational tractability, the model is built upon several key assumptions and strategic simplifications that are fundamental to its design and the interpretation of its results.

1. **Closed Ecological System:** The simulation domain is treated as a closed system using periodic boundary conditions, which effectively create a toroidal space. This design eliminates artificial edge effects that could bias agent behavior near the boundaries and ensures that all population changes arise strictly from the model's internal dynamics—birth, natural mortality, and fishing—with no external migration.
2. **Single-Species Focus:** The model simplifies the broader marine ecosystem by concentrating on the dynamics of a single fish species. Interspecific interactions such as predation or competition with other species are not explicitly simulated. Instead, the influence of the wider food web is implicitly represented through the dynamic habitat quality layer, which serves as a proxy for food availability and overall ecosystem health.
3. **Asymmetrical Agent Information:** A critical asymmetry in information availability is assumed between the agent types. Fish agents are endowed with perfect and instantaneous knowledge of the environmental conditions in their immediate neighboring cells, allowing them to effectively navigate environmental gradients. Conversely, fisher agents operate under bounded rationality with imperfect information, their knowledge being limited to their vessel's local sonar range and a finite memory of their own past successful catches, realistically modeling the search problem they face.

4. **Static Economic Environment:** The economic parameters that drive fisher behavior are held constant throughout the simulation. Key variables such as the market price for fish, fuel costs, and fixed operational costs do not fluctuate. This simplification isolates the analysis of fisher behavior from external market volatility, allowing the model to focus on how decisions are shaped primarily by resource availability and operational constraints.
5. **Homogeneity of Agents:** Agents within their respective classes are considered homogeneous in their intrinsic capabilities. All fisher agents are endowed with identical vessel characteristics, technical efficiency, and decision-making logic. Similarly, all fish share the same biological parameters for growth and mortality based on their age. This ensures that any observed differences in individual performance, such as vessel profitability, emerge directly from their unique spatial interactions and stochastic experiences rather than from pre-defined variations in skill or quality.

3.5 Simulation Analysis, Metrics, and Outputs

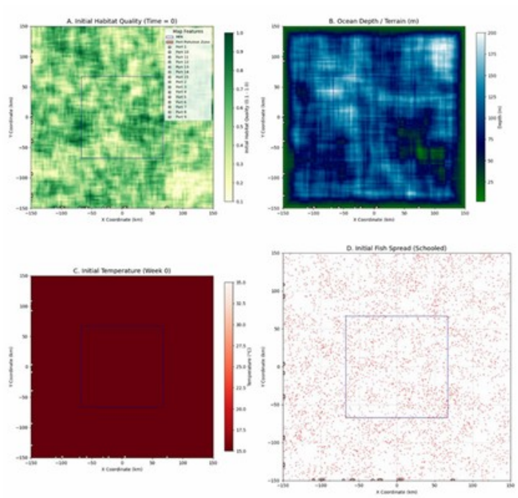
The simulation is instrumented to capture a rich dataset, enabling a multi-scalar analysis of the system's dynamics and the performance of the MPA.

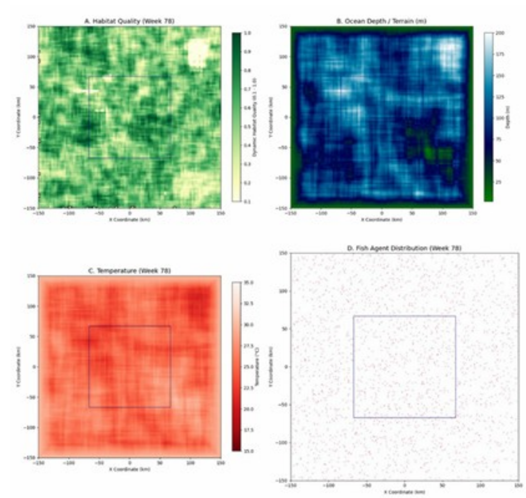
- Qualitative Visual Outputs: The model generates a suite of graphical outputs for qualitative analysis and visual interpretation. These include an initial multi-panel plot displaying the starting conditions of all environmental layers and the initial schooled distribution of fish. This is followed by periodic snapshots during the simulation's run to visualize the co-evolution of the habitat and the fish population. Finally, a series of summary plots are generated at the simulation's conclusion, including a final spatial distribution map showing fish, fisher activity, and ports, and a comprehensive time-series plot illustrating the longitudinal trends of key system-level metrics.
- Quantitative Data Outputs: For rigorous quantitative analysis, the model produces two detailed tabular datasets. The first provides an aggregated, system-level view of the simulation on a weekly basis, containing records of the total fish population disaggregated by life stage, the total fleet-wide catch, and the total net profit realized

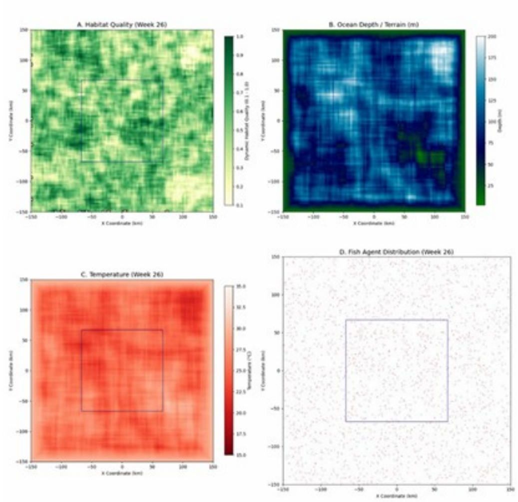
by the entire fleet. The second dataset provides a highly granular, disaggregated view of the fleet's economic performance. It contains records for each individual vessel's trip for a given week, detailing its catch quantity, gross revenue, fixed operational costs, variable fuel costs, and the resulting net profit or loss. This detailed dataset allows for in-depth analysis of the economic heterogeneity, behavioral patterns, and financial viability of individual actors within the non-cooperative fleet.

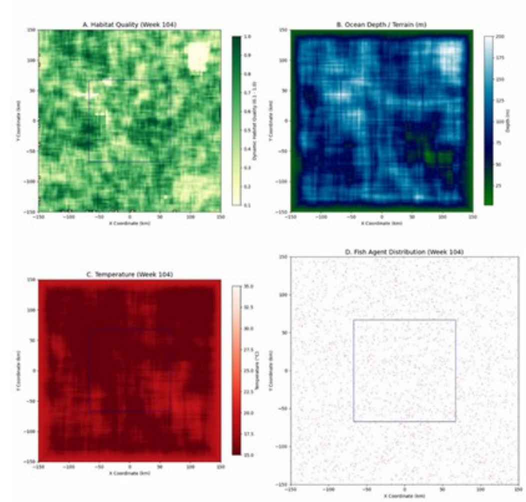
CHAPTER 4

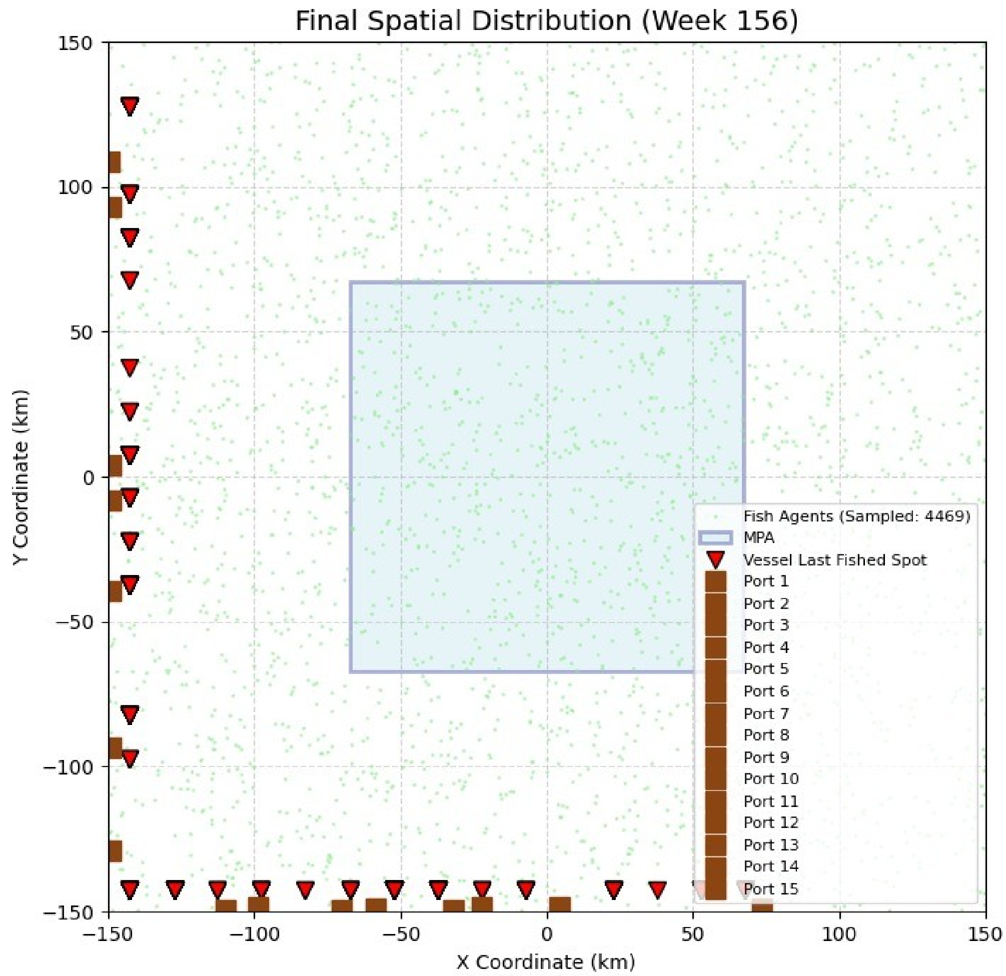
RESULTS



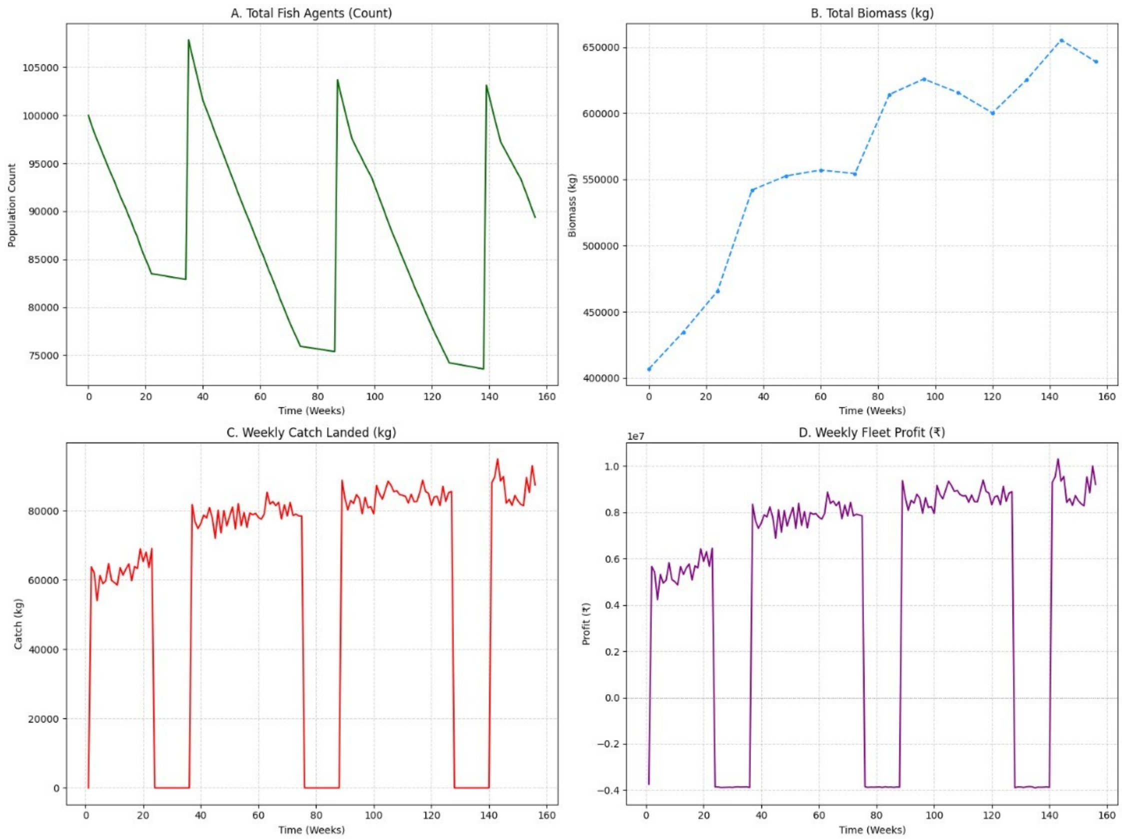








Final Spatial Distribution (Week 156)
Y Coordinate (km)
X Coordinate (km)
150
100
50
0
−50
−100
−150
−150
−100
−50
0
50
100
150
Fish Agents (Sampled: 4469)
MPA
Vessel Last Fished Spot
Port 1
Port 2
Port 3
Port 4
Port 5
Port 6
Port 7
Port 8
Port 9
Port 10
Port 11
Port 12
Port 13
Port 14
Port 15



The simulation was conducted over a period of 156 weeks (approximately 3 years) to analyze the interplay between fishery dynamics, fleet behavior, and the ecological impact of a central Marine Protected Area (MPA). The findings are presented through time-series analyses of key biological and economic indicators, spatial distribution plots, and snapshots of the habitat conditions.

4.1 Temporal Dynamics of the Fishery

The overall temporal trends for the fish population and the fishing fleet are summarized in Figure 1. The total fish agent population count (Figure 1A) displays a distinct cyclical pattern, with sharp annual recovery events followed by periods of decline. Despite these fluctuations, the total biomass of the fish stock shows a clear and sustained increasing trend over the 156-week period, rising from an initial value of approximately 400,000 kg to a final value of nearly 650,000 kg (Figure 1B). This suggests an increase in the average size and weight of individual fish agents within the system.

The economic outputs of the fishery reflect a regulated fishing season. Both the weekly catch landed (Figure 1C) and the weekly fleet profit (Figure 1D) exhibit clear on- and off-seasons. Each year includes a period of approximately 12-14 weeks where fishing activity ceases, catch drops to zero, and the fleet incurs a net loss, likely due to fixed operational costs. During the active fishing seasons, both catch and profit levels are relatively stable, with a slight upward trend year-on-year, mirroring the growth in total biomass.

4.2 Spatial Distribution and MPA Efficacy

The spatial configuration of the fishery at the conclusion of the simulation (Week 156) is shown in Figure 2. Fishing activity, indicated by the last-fished locations of the vessels, is confined entirely to the areas outside the designated MPA. A notable concentration of fishing effort is observed along the boundaries of the MPA, demonstrating the fleet's adaptation to the spatial closure.

The MPA's effectiveness as a refuge is evident in the fish agent distribution. At Week 156, the density of fish agents is visibly higher inside the MPA than in the surrounding fished areas (Figure 2). This "spillover effect," where the protected population replenishes adjacent areas, is a key outcome. Analysis of the fish distribution at various time points (e.g., Weeks 26, 78, and 156, as seen in Figure 3D, Figure 4D, and Figure 5D respectively) confirms a gradual accumulation of the fish population within the MPA's borders over the course of the simulation.

4.3 Environmental Conditions

Snapshots of the simulated marine environment were captured at multiple intervals (Figures 3, 4, and 5 provide examples from Weeks 26, 78, and 156). The ocean depth and terrain (Subplot B in all habitat figures) and the sea temperature (Subplot C) were static variables throughout the simulation run. In contrast, habitat quality (Subplot A) was dynamic, showing subtle spatial and temporal variations over the 156 weeks, though no drastic system-wide changes were observed. The distribution of fish agents (Subplot D) appears to correlate with higher-quality habitats, particularly within the MPA where fishing pressure is absent.

The dominant paradigms in fishery modeling can be broadly categorized as "top-down" (equation-based) and "bottom-up" (agent-based). Top-down models, epitomized by frameworks such as **marlin**, excel at simulating system-level dynamics for large-scale, strategic inquiries. They operate by representing entire fish populations and fishing fleets as aggregate entities distributed across a spatial grid. Instead of tracking individual organisms, these models use deterministic equations to govern population-level processes like growth, mortality, and recruitment. The fishing fleet is often modeled as a monolithic unit, with its spatial effort allocated based on rules derived from the aggregate profitability of different areas in a preceding time step. This top-down abstraction is computationally powerful, enabling the rapid evaluation of broad policy scenarios. However, this efficiency is achieved by sacrificing granular detail; the heterogeneity within a fleet, the proactive and often unpredictable decisions of individual captains, and the fine-scale spatial clustering of fish are necessarily smoothed over.

In contrast, the "bottom-up" paradigm of agent-based modeling (ABM) builds system-level understanding from the ground up by simulating the actions and interactions of individual, autonomous agents. This approach allows complex, system-wide patterns to emerge organically from the simple rules governing each agent. Models like **POSEIDON** and the Corsican fishery model leverage this paradigm to represent the human component of the fishery with greater fidelity. An ABM can capture the diversity of motivations, knowledge, and constraints that exist within a real-world fishing fleet, providing a richer understanding of how regulations might be perceived and acted upon at the individual level. Our model fundamentally aligns with this bottom-up philosophy, asserting that for many tactical management questions, these individual-level dynamics are not noise to be abstracted away, but are in fact the primary engine of system behavior.

Hybrid models seek to blend the strengths of both paradigms, leveraging the efficiency of top-down equations for well-understood, systemic processes while using the bottom-up power of agents for behaviors that are complex, adaptive, and emergent. The Corsican fishery model employs a hybrid structure by using a Cellular Automata Model (CAM)—a grid-based, rule-driven system—for the environment, coupled with an ABM for its fishermen. Our model advances this hybrid philosophy by creating a more deeply integrated and behaviorally complex framework. We use a bottom-up, agent-based approach for the components with the highest behavioral uncertainty—the fish and the fishers—allowing for emergent phenomena like schooling and adaptive fishing. Simultaneously, we use top-down, equation-based rules for systemic processes that are biologically well-established and computationally expensive to simulate at the agent level, such as individual fish growth (Von

Bertalanffy) and population-wide recruitment (Beverton-Holt). This creates a powerful feedback loop where the collective actions of individual agents (e.g., their total biomass) provide the aggregate data needed to parameterize the global equations, and the outputs of those equations in turn define the environmental conditions that shape future agent decisions. This strategic synthesis allows us to achieve a high degree of realism without incurring the prohibitive computational cost of a purely bottom-up simulation of every biological process.

Our model's architecture offers a distinct alternative to equation-based frameworks like **marlin**. While **marlin** is a highly effective tool for strategic, high-level analysis, its abstraction of individual behavior limits its applicability for certain tactical questions. For instance, **marlin** simulates fish movement as a generalized process of diffusion and taxis across a grid. In contrast, our model simulates thousands of individual fish agents making autonomous movement decisions. This allows for the emergent formation of dynamic schools and aggregations in favorable habitats—a critical ecological feature that directly dictates the success and spatial pattern of fishing operations. The resource landscape in our model is not a static grid to be acted upon; it is a living, shifting mosaic created by the collective behavior of thousands of individual fish agents. Furthermore, we model each vessel as an autonomous, profit-maximizing agent with memory and exploratory capabilities. This stands in contrast to **marlin**'s fleet, which reacts based on historical aggregate data. Our fisher agents proactively shift their effort in response to localized stock depletion and economic returns, capturing the adaptive nature of fishing pressure more realistically. The trade-off is one of computational focus versus ecological fidelity. **marlin** is optimized for rapid, broad-scale assessments. Our model, while more computationally intensive, provides a higher-fidelity representation of the local processes that are often the specific target of management interventions, such as regulations designed to protect spawning aggregations or reduce bycatch in mixed fisheries.

POSEIDON represents the frontier of agent-based modeling with its sophisticated focus on the complexities of human behavior. Its central innovation is the "explore-exploit-imitate" (EEI) algorithm, which allows individual fisher agents to learn from each other through social networks, enabling the simulation of emergent fleet-level behaviors. The parameters in **POSEIDON** are thus heavily geared towards agent behavior and policy design, including attributes like gear efficiency, the structure of social networks, and variables defining policy instruments like quota levels or MPA boundaries. While this provides a powerful lens into the socio-economic dynamics of a fishery, the ecological component is simplified to serve as a backdrop for human action; fish are represented as an aggregated biomass resource within a grid. Our model complements this by architecting a more balanced socio-ecological system. We elevate the ecological component by simulating fish as individual agents with their own complex behaviors. This allows us to investigate the direct "predator-prey" dynamic between individual vessels and the mobile schools of fish they are actively searching for, a level of detail beyond the scope of a model with an aggregated fish stock.

This architectural difference is directly reflected in their computational performance. The time complexity of **POSEIDON** is approximately $O(F^2 + P)$, where F is the number of fisher agents. The quadratic scaling with fleet size (F^2) is a direct consequence of its social learning algorithm; the "imitate" step requires each agent to compare its performance against others, a process that becomes exponentially more demanding as the fleet grows. In contrast, the time complexity of our model is approximately $O(N_{fish} \times F_{fishers} + G_{dim}^2)$. The key term is the multiplicative relationship between the number of fish and fisher agents, which represents the fundamental ecological interaction of searching and harvesting. Our model avoids a quadratic dependence on fleet size, making it potentially better suited for simulating large, industrial fisheries where individual, profit-driven search may be a more dominant behavior than direct social imitation.

The Corsican fishery model provides a valuable benchmark for a computationally efficient, rule-based hybrid approach. Its parameters are grounded in specific bioeconomic theory and local fishery rules, including fisher catchability (q), logistic growth rate (r), carrying capacity (k), and fixed costs. Its agents are not adaptive learners but follow a prescribed, deterministic routine governed by a finite state machine. This makes the model exceptionally fast, with a time complexity of $O(F \quad + P)$ that scales linearly with the number of environmental patches and fisher agents. However, this efficiency comes at the cost of behavioral plasticity. The model's agents cannot adapt to novel conditions that fall outside their programmed routine. Our model offers a significant advantage here. Our fisher agents are profit-driven, adaptive entities that can abandon depleted fishing spots and explore new territories in response to changing economic incentives. The parameters in our model, such as sonar radius, maximum fuel range and fixed operational cost, are designed to create a set of constraints and drivers that enable, rather than prescribe, complex adaptive behavior. While the Corsican model is excellent for simulating the stability of an existing, well-understood system, our model is explicitly designed as a predictive tool for exploring a system's resilience and its response to disruption and change. The higher computational complexity of our framework is a direct and necessary trade-off for this enhanced behavioral realism and predictive power.

In synthesizing these comparisons, our proposed architecture occupies a unique and powerful position. It integrates the computational efficiency of top-down ecological equations with the high-fidelity, emergent realism of a dual-population ABM, providing a uniquely balanced tool for exploring the intricate feedback loops between marine life and human enterprise. It avoids the pure abstraction of equation-based models like **marlin**, provides a more detailed and dynamic ecological foundation than behavior-focused ABMs like **POSEIDON**, and offers far greater behavioral flexibility and predictive power than rule-based models like the Corsican example. This hybrid architecture provides a high-fidelity tool uniquely positioned to address the complex, tactical, and increasingly urgent questions at the heart of modern fisheries management. By simulating both ecological and economic agents with a high degree of autonomy, we can better understand and predict how the finely woven fabric of a marine ecosystem responds to the pressures of human activity.

4.4 Calculating the Depth of a Single Cell

The following is a dry run focusing only on how the combined noise value for a *single, imaginary cell* is calculated and then converted into its final depth.

The Scenario

Let's imagine we are calculating the final depth for just one cell in the 100x100 grid. After the initial smoothing (convolution), let's assume this specific cell has the following values from each of the three noise layers:

- Value from Layer 1 (Small Roughness): *0.70*
- Value from Layer 2 (Medium Features): *0.50*
- Value from Layer 3 (Large Features): *0.20*

Step 1: Weighted Sum

First, we combine these three values using the specified weights (20% for Layer 1, 40% for Layer 2, 40% for Layer 3).

- *Formula:* (Value_L1 * 0.20) + (Value_L2 * 0.40) + (Value_L3 * 0.40)
- *Calculation:* (0.70 * 0.20) + (0.50 * 0.40) + (0.20 * 0.40) = 0.14 + 0.20 + 0.08 = 0.42

After this step, our cell's raw, combined noise value is *0.42*.

Step 2: Normalization

This step requires knowing the minimum and maximum raw values across the entire grid. Since we are only looking at one cell, let's assume that after calculating the weighted sum for all 10,000 cells, we found:

- The lowest value anywhere on the grid was *0.15*.
- The highest value anywhere on the grid was *0.85*.

Now, we normalize our cell's value (0.42) to fit within this range.

- *Formula:* (OurValue - GridMin) / (GridMax - GridMin)
- *Calculation:* (0.42 - 0.15) / (0.85 - 0.15) = 0.27 / 0.70 ≈ 0.3857

Our cell's normalized value is now *0.3857*. This number is guaranteed to be between 0.0 and 1.0.

Step 3: Scaling to Depth

Finally, we convert this normalized [0, 1] value into a real-world depth between 5m (the minimum) and 200m (the maximum).

- First, find the total depth range: 200m - 5m = 195m.
- *Formula:* MinDepth + (NormalizedValue * TotalDepthRange)
- *Calculation:* 5 + (0.3857 * 195) = 5 + 75.21 = 80.21

The final calculated depth for our imaginary cell is *80.21 meters*. This entire process is performed for every cell in the grid simultaneously using efficient array operations.

CHAPTER 5

CONCLUSION AND FUTURE WORK

4.1 CONCLUSION

. In conclusion, our research has successfully developed and demonstrated a novel hybrid simulation architecture that advances the field of computational fisheries science. By integrating a bottom-up, dual-agent-based model for both individual fish and fishers with top-down, equation-based components for systemic ecological processes, we have created a uniquely balanced and powerful tool. This model transcends the limitations of singular paradigms; it provides greater bio-behavioral fidelity than top-down frameworks like **marlin** and a more holistic socio-ecological representation than behavior-focused ABMs like **POSEIDON** or rule-based systems like the Corsican fishery model. Our approach allows for the emergence of critical, fine-scale phenomena such as fish schooling and adaptive, profit-driven fishing pressure, while remaining grounded in established ecological principles for growth and recruitment, ensuring long-term population stability.

The primary implication of this work is the creation of a robust and flexible platform for Management Strategy Evaluation (MSE), particularly for tactical questions involving complex spatial dynamics and adaptive fisher responses. The model is uniquely positioned to explore how marine socio-ecological systems react to novel disruptions, such as climate-induced habitat shifts, market volatility, or the implementation of new, spatially explicit regulations. While this high-fidelity approach entails a significant computational cost, driven primarily by the direct interaction between numerous fish and fisher agents, we contend that this is a necessary trade-off for achieving a realistic depiction of the core predator-prey dynamic. Future work will focus on optimizing these interaction algorithms, calibrating the model with rich empirical datasets, and extending its capabilities to include multi-species dynamics and more sophisticated human decision-making heuristics, further enhancing its value as a predictive tool for sustainable marine resource management.

4.2 FUTURE WORK

Building upon the robust hybrid architecture established in this study, our future research will focus on expanding the model's capabilities in four key areas to enhance its realism and utility as a decision-support tool for marine resource management. Our immediate goal is to advance the optimization framework to address the multi-faceted nature of policy trade-offs more explicitly. We will move beyond single-objective optimization to incorporate a multi-objective approach, allowing for the simultaneous consideration of competing goals such as maximizing fishery yield, protecting vulnerable species, ensuring equitable economic returns across different fleet segments, and minimizing ecological disruption. This will involve developing a Pareto front analysis to visually represent the trade-offs between these objectives, enabling stakeholders to identify policies that represent an optimal balance rather than an extreme solution. Furthermore, we intend to explore the design of dynamic or adaptive MPAs, whose boundaries and regulations can evolve in response to simulated environmental changes, such as climate-driven species migrations, providing a tool to design more resilient, future-proof conservation strategies.

Second, we will significantly enhance the sophistication of the human behavioral component by modeling the impact of cooperative and collective action among fisher agents. The current model treats fishers as independent, profit-maximizing entities, a simplification that overlooks the powerful role of social dynamics in real-world fisheries. Our next step is to implement mechanisms for cooperation, such as information sharing about productive fishing grounds, the formation of co-management bodies for self-policing, and collective agreements to avoid over-exploitation of shared resources. By incorporating these social network dynamics, our model will be able to investigate critical questions about the conditions under which cooperation emerges and how it influences the effectiveness of top-down regulations like MPAs. This will allow us to simulate scenarios where community-led initiatives can complement or even outperform traditional management, providing valuable insights for designing more equitable and effective governance structures.

Third, to achieve a more holistic understanding of MPA impacts, we will expand the model’s ecological scope by integrating a multi-species food-web model. The current framework, while detailed in its population dynamics, does not account for the trophic interactions between species. By incorporating predator-prey relationships and resource competition, we can move from a multi-species simulation to a true ecosystem model. This enhancement will enable the investigation of potential indirect effects of MPA implementation, such as trophic cascades, where the protection of a top predator leads to cascading changes down the food web. Assessing these broader ecosystem-level interactions is crucial for avoiding unintended consequences and ensuring that conservation efforts support the health and stability of the entire marine environment, not just the target species.

Finally, to bridge the gap between complex computational science and practical application, our ultimate objective is to develop an interactive decision-support interface. This tool will translate the rich, multi-dimensional outputs of our simulation into an accessible and intuitive format for non-expert users, including policymakers, fishery managers, and community stakeholders. The interface will feature a map-based visualization where users can design their own MPA networks, adjust policy parameters, and run simulations to observe the predicted ecological and economic consequences in near real-time. By empowering stakeholders to explore scenarios and understand the trade-offs inherent in their decisions, we aim to transform our model from a purely academic research tool into a collaborative platform for evidence-based, participatory marine spatial planning.

REFERENCES

[1] Scoville, G., Chapman, J., Amironesei, R., & Boettiger, C. (2021). *Ecological reinforcement learning*. arXiv preprint arXiv:2104.05888.

[2] Combes, V., Gomei, M., El-Haddad, C. F., O'Brien, A., Abdulla, A., Al-Itr, M., ... & Micheli, F. (2021). Systematic conservation planning at an ocean basin scale: Identifying a portfolio of priority areas for conservation in the Mediterranean Sea. *Frontiers in Marine Science*, *8*, 742152. https://doi.org/10.3389/fmars.2021.742152

[3] Ovando, D., Kaplan, I. C., Marshall, K. N., Essington, T. E., & Levin, P. S. (2021). *marlin: a simulation framework for spatially-explicit, multi-species, multi-fleet fisheries models to inform marine spatial management*. NOAA Technical Memorandum NMFS-NWFSC-168. https://doi.org/10.25923/6j5b-s117

[4] Thanassekos, S., & Scheld, A. M. (2021). Simulating the effects of environmental and market variability on fishing industry structure. *Ecological Economics*, *188*, 107129. https://doi.org/10.1016/j.ecolecon.2021.107129

[5] Pérez, J. G., Tizón, L. R. G., Carral, L., & Fraguela, J. Á. F. (2021). Exact zoning optimization model for marine spatial planning. *Frontiers in Marine Science*, *8*, 753820. https://doi.org/10.3389/fmars.2021.753820

[6] Poulton, C. R., Sethi, S. A., Ellner, S. P., & Smeltz, T. S. (2022). Optimal dynamic spatial closures can improve fishery yield and reduce fishing-induced habitat damage. *Canadian Journal of Fisheries and Aquatic Sciences*, *79*(6), 947-960. https://doi.org/10.1139/cjfas-2021-0205

[7] Fourchault, Q., Visalli, M. E., O'Leary, B. C., Tuda, A. O., Munga, C. N., Roberts, C. M., & Harris, L. R. (2023). Affordable protection of high seas biodiversity through cross-sectoral spatial planning. *One Earth*, *6*(8), 1058-1070. https://doi.org/10.1016/j.oneear.2023.07.010

[8] Álvarez-Romero, J. G., Munguia-Vega, A., Beger, M., Mancha-Cisneros, M. D. M., Suárez-Castillo, A. N., Garcés-García, L. C., ... & Ban, N. C. (2023). Marine spatial planning for connectivity and conservation through ecological corridors between marine protected areas and other effective area-based conservation measures. *Frontiers in Marine Science*, *10*, 1088118. https://doi.org/10.3389/fmars.2023.1088118

[9] Roshani, M., Salmanmahiny, A., Mirkarimi, S. H., & Pourmanafi, S. (2024). Spatial planning of marine protected areas in the Southern Caspian Sea: A comparison of MCE

and simulated annealing algorithm results. *Journal of Marine Science and Engineering*, *12*(1), 161. https://doi.org/10.3390/jmse12010161

[10] Hanson, J. O., Schuster, R., Strimas-Mackey, M., & Bennett, J. R. (2024). Spatial conservation planning: Proposing clustering methods to solve large prioritization problems. *Methods in Ecology and Evolution*, *15*(2), 246-258. https://doi.org/10.1111/2041-210X.14258

[11] Equihua, J., Beckmann, M., & Seppelt, R. (2024). Connectivity conservation planning through deep reinforcement learning. *Methods in Ecology and Evolution*, *15*(1), 133-146. https://doi.org/10.1111/2041-210X.14247

[12] Brunel, C., Omer, J., Gicquel, C., & Lanco, G. (2024). Designing compact, connected and gap-free reserves with systematic reserve site selection models. *Applied Mathematical Modelling*, *125*, 788-806. https://doi.org/10.1016/j.apm.2023.09.020

[13] Hanson, J. O., Schuster, R., Strimas-Mackey, M., & Bennett, J. R. (2023). Systematic conservation prioritization with the prioritizr R package. *Conservation Biology*, *37*(1), e13999. https://doi.org/10.1111/cobi.13999

[14] Magris, R. A., Trebilco, R., & Williams, S. L. (2024). prior3D: An R package for three-dimensional conservation prioritization. *Ecological Modelling*, *490*, 110645. https://doi.org/10.1016/j.ecolmodel.2024.110645

[15] Ovando, D. (2024). Predicted effects of marine protected areas on conservation and catches are sensitive to model structure. *Theoretical Ecology*. https://doi.org/10.1007/s12080-024-00591-2

[16] Soulié, J. C., & Thébaud, O. (2006). Simulating fishery dynamics by combining empirical data and agent-based modeling: The case of the Channel and North Sea bass fishery. *Ecological Economics*, *59*(2), 169-183. https://doi.org/10.1016/j.ecolecon.2005.10.007

[17] DePiper, G. S., Gaichas, S. K., Lucey, S. M., Pinto da Silva, P., Olson, J., & Adams, C. F. (2017). Modeling consequences of spatial closures for offshore energy development. *Ecological Applications*, *27*(4), 1143-1157. https://doi.org/10.1002/eap.1511

[18] Janssen, M. A., Anderies, J. M., & Cifdaloz, O. (2010). Effects of cooperation and different characteristics of marine protected areas on fish stocks. *Ecological Economics*, *69*(4), 833-842. https://doi.org/10.1016/j.ecolecon.2009.11.002

[19] Travers, M., Shin, Y. J., Jennings, S., & Cury, P. (2007). Towards a standard approach for representing food webs in ecosystem models. *ICES Journal of Marine Science*, *64*(4), 689-696. https://doi.org/10.1093/icesjms/fsm010

[20] Cabral, R. B., Blasiak, R., Boustany, A., Crona, B., de souza, E. N., Gjerde, K. M., ... & Halpern, B. S. (2024). Global expansion of marine protected areas and the risk of fishing effort redistribution. *Proceedings of the National Academy of Sciences*, *121*(28), e2400592121. https://doi.org/10.1073/pnas.2400592121 *em*of m

[21] Bailey, R. M., Carrella, E., Axtell, R., Burgess, M. G., Cabral, R. B., Drexler, M., ... & Saul, S. (2019). A computational approach to managing coupled human-environmental systems: The POSEIDON model of ocean fisheries. *Sustainability Science, 14(2), 259-275*. https://doi.org/10.1007/s11625-018-0579-9

[22] Innocenti, E., Idda, C., Prunetti, D., & Gonsolin, P. R. (2020). Agent-based modelling of a small-scale fishery in Corsica. *Proceedings of the 10th International Workshop on Simulation for Energy, Sustainable Development & Environment, 005.* https://doi.org/10.46354/i3m.2022.sesde.005